a low latency, will be a lot more feasible. The uniformity of the connection will be maintained, hence ensuring that your experience is seamless.

Mission-critical services. Self-driving cars require real-time connection with the servers, which can be possible with 5G. A real-time connection also helps in making a better decision and much more access, especially in the robotics sector. In healthcare sector, 5G can make it easier to conduct medical procedures remotely due to its ultra-low latency and strong security.

Internet of Things (IoT) devices. A lot of IoT devices demand good connectivity to function properly and provide users with a hassle-free experience, which can easily be taken care of with 5G. It will improve their overall coverage at a very affordable cost.

How the mobile network works

Mobile devices are kind of IoT devices. All mobile devices are connected to various cell towers that are known as base stations. A group of such towers forms a radio access network and remains connected to the core network.

The core network is responsible for client profiling. It provides access to various services, such as subscriber trunk dialing (STD) calls and data services. Signals from the towers are sent to the core network and the core network is responsible for all the further tasks, such as the execution of the final service. It can be any kind of service, such as an internet connection. The core network, thus, acts as a gateway between the radio access network and the final services (see Fig. 1).

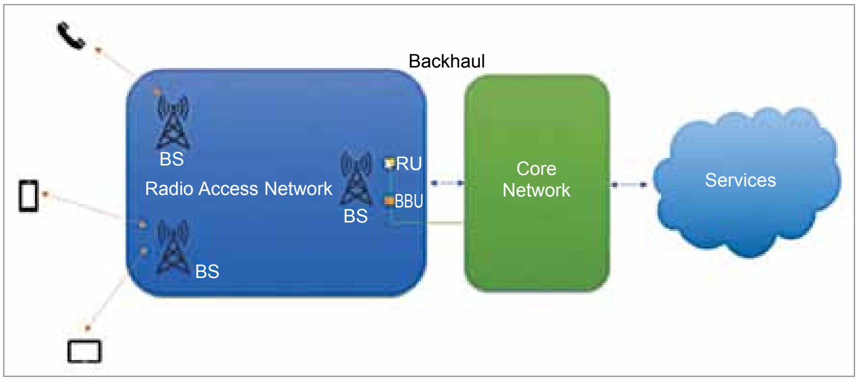

Fig. 1: Mobile network

Traditional vs open RAN

Before the introduction of 5G, the base station had mainly two units—the radio unit (RU) or the signal transmitting unit and the baseband unit (BBU) for processing of the signals. RU receives the signals and sends to BBU. Only after that the signals are sent to the core network.

Earlier on, a single vendor used to provide all the units and the connection between them was also deployed by that vendor only. There used to be a very tightly coupled system, which made it hard to change any component. We use software to run the hardware, but earlier on that software was very tightly integrated inside the hardware, which presented a challenge to change the software as well.

So, it was difficult to upgrade the infrastructure besides full dependency on a single vendor. If one wanted to shift to a different vendor, the entire hardware and the software had to be changed, which was never convenient.

Open RAN (O-RAN) makes both hardware and software follow a set of rules so that any vendor can come up with their own solutions. It allows hardware and software from different vendors to work together.

Servers can now independently implement virtualisation due to this desegregation. You do not need a physical server for the mandate. O-RAN even makes cloud solutions possible. It lets one virtualised baseband unit support multiple radios instead of needing a proprietary or a physical unit at every cell site. It is much more flexible and cost-effective at the same time.

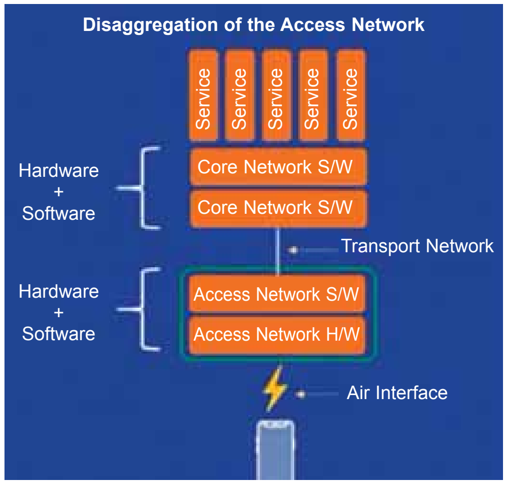

Fig. 2: Typical mobile architecture

Benefits of open O-RAN

Hardware and software disaggregation. The hardware and the software are independent of each other.

Intelligence and automation. It makes open management and orchestration more feasible, since everything is quite open and there are standard rules to do things. It uses external artificial intelligence and machine learning capabilities.

Open internal RAN interfaces. O-RAN lowers layer split, which basically splits the previous single unit to multiple units and mobilises them and makes a multi-vendor solution.

Rapid innovation. It has improved the innovation pace significantly due to the increased competi-